

Quote of the day: “*Induction makes you feel guilty for getting something out of nothing, and it is artificial, but it is one of the greatest ideas of civilization.*” – Herbert S. Wilf

Quote of the day 2: “*If one proves the equality of two numbers a and b by showing first that $a \leq b$ and then that $a \geq b$, it is unfair: one should instead show that they are really equal by disclosing the inner ground for their equality.*” – Emmy Noether

Main reference: Clive Newstead, *An Infinite Descent into Pure Mathematics*

1. PROOF BY INDUCTION

The main idea of proof by induction is the following, which uses some very special but intuitive properties of the natural numbers, which we will take for granted.

Big Idea 1.0.1. Suppose we want to climb a set of stairs indexed by the natural numbers, i.e., we have steps $0, 1, 2, 3, 4, \dots$. Since we know how to get from the n th step to the $(n + 1)$ th step, as long as we can get onto the 0th step, we can climb all of the stairs.

Big Idea 1.0.2. Dominoes: if we set up a line of dominoes such that each domino knocks down the next in line, to knock down all of the dominoes, it suffices to knock over the first one!

Theorem 1.0.3. Suppose we want to show that a logical formula $p(n)$, with $n \in \mathbb{N}$, is true for all $n \in \mathbb{N}$. Suppose further that we know the following two statements are true:

- (i) for all $n \in \mathbb{N}$, $p(n)$ is true implies that $p(n + 1)$ is true;
- (ii) $p(0)$ is true.

Then $p(n)$ is true for all $n \in \mathbb{N}$!

Proof Strategy 1.0.4. So, suppose we have a logical formula $p(n)$ where $n \in \mathbb{N}$. To prove that $p(n)$ is true, we need to prove the following two statements, called the **base case** and **induction step**:

Base Case: $p(0)$ is true.

Induction step: Assume that $p(n)$ is true, and prove that $p(n + 1)$ is true.

The assumption that $p(n)$ is true is called the **induction hypothesis**.

Remark 1.0.5. Sometimes, it makes sense to “start” induction at a natural number other than 0, most commonly 1 or 2. In this case, an induction proof with base case starting at 1 would prove a statement is true for all $n \geq 1$.

Proposition 1.0.6. Let $n \in \mathbb{N}$. Then

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Proof. We proceed by induction on $n \geq 1$.

Base Case: We will show the statement is true for $n = 1$. Indeed,

$$\sum_{k=1}^1 1 = 1 = \frac{1(1+1)}{2} = 2/2 = 1$$

Induction Step: Under the assumption that the statement is true for n , we show that the statement is true for $n + 1$. So, suppose that

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

We want to show that

$$\sum_{k=1}^{n+1} k = \frac{(n+1)(n+1+1)}{2} = \frac{(n+1)(n+2)}{2}.$$

We have

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \left[\sum_{k=1}^n k \right] + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) && \text{(induction hypothesis)} \\ &= \frac{n(n+1)}{2} + \frac{2(n+1)}{2} \\ &= \frac{(n+2)(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

Hence we have proved the base case and the induction step, so the statement is true for all $n \in \mathbb{N}$. □

A second proof of Proposition 1.0.6. We consider the sum

$$\begin{array}{c} 1 + 2 + \dots + n - 1 + n \\ n + n - 1 + \dots + 2 + 1 \end{array}$$

where we want to add the top row to the bottom row. Notice that the top row and the bottom row both add up to $\sum_{k=1}^n k$, so we know that the result of this addition is $2 \sum_{k=1}^n k$. The key point is that we can add in whatever order we want! So we can add vertically first:

$$\begin{array}{ccccccc} 1 & + & 2 & + & \dots & + & n-1 & + & n \\ n & + & n-1 & + & \dots & + & 2 & + & 1, \end{array}$$

so we know that sum must also be equal to $n(n+1)$, since we are adding the number $n+1$ to itself n times. Therefore, we have

$$2 \left[\sum_{k=1}^n k \right] = n(n+1),$$

so dividing by two on both sides, we obtain the desired result. □

Proposition 1.0.7. *For all $n \in \mathbb{N}$, $n^3 - n$ is divisible by 3.*

Proof. We proceed by induction on $n \geq 0$.

Base Case: We need to prove that $0^3 - 0$ is divisible by 3. Indeed, $0^3 - 0 = 0 = 3 \cdot 0$, so 3 divides $0^3 - 0$.

Induction Step: Under the assumption that $n^3 - n$ is divisible by 3, we show that $(n + 1)^3 - (n + 1)$ is divisible by 3. By the induction hypothesis, we know that $n^3 - n = 3k$ for some $k \in \mathbb{Z}$. To show that $(n + 1)^3 - (n + 1)$ is divisible by 3, we must show that there exists some $m \in \mathbb{Z}$ such that $(n + 1)^3 - (n + 1) = 3m$. We have

$$\begin{aligned}(n + 1)^3 - (n + 1) &= (n^3 + 3n^2 + 3n + 1) - n - 1 \\&= n^3 - n + 3n^2 + 3n + 1 - 1 \\&= n^3 - n + 3n^2 + 3n \\&= 3k + 3n^2 + 3n && \text{(induction hypothesis)} \\&= 3(k + n^2 + n).\end{aligned}$$

Hence $(n + 1)^3 - (n + 1)$ is divisible by 3. □

2. IF WE HAVE TIME...A FAILED INDUCTION PROOF

Example 2.0.1. The following argument supposedly proves that every horse is the same color.

- **Base Case:** Suppose there is just one horse. This horse is the same color as itself, so the base case is true.
- **Induction Step:** Suppose that every collection of n horses is the same color. Let X be a set of $n + 1$ horses. Removing the first horse from X , we see that the last n horses are the same color by the induction hypothesis. Removing the last horse from X , we see that the first n horses are the same color. Hence all the horses in X are the same color.

By induction, we have proved that all horses are the same color. What's wrong with this proof?

Remark 2.0.2. The "horses" version of this paradox was presented in 1961 in a satirical article by Joel E. Cohen. It was stated as a lemma, which in particular allowed the author to "prove" that Alexander the Great did not exist, and he had an infinite number of limbs.